

Obsidian Protocol Whitepaper

The gold standard on Solana

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Entity: BLKW3B Inc. • Delaware C-Corporation • EIN 41-5179913

Contact: blckw3b@gmail.com • w3bs.fun

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This document describes the architecture, design principles, and ecosystem positioning of Obsidian Protocol, a tokenized precious metals infrastructure layer built natively on the Solana blockchain by BLKW3B Inc.

ABSTRACT

The global precious metals market represents trillions of dollars in held value across gold, silver, and alternative sound money instruments. Despite this scale, the market has no true production-ready on-chain infrastructure for physical redemption. No named custodian. No burn-to-redeem mechanic. No institutional distribution. Obsidian Protocol solves this by issuing five SPL Token 2022 assets on Solana, each backed 1:1 by physical metal held in insured vault custody via United Precious Metals Association (UPMA) servicing Alpine Gold Exchange (AGX). Reserve ratios are attested daily via Light Protocol ZK compression proofs published on-chain. Physical redemption is available to any token holder. Protocol revenue is captured automatically via SPL Token 2022 transfer hooks on every transaction. An x402 payment gateway makes the protocol accessible to AI agents and autonomous systems through pay-per-call endpoints. Obsidian Protocol is the Stripe and Venmo for physical precious metals, built on the fastest and cheapest settlement layer in the world.

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1. THE PROBLEM

Physical precious metals are the oldest store of value in human history. Gold has been used as money for over 5,000+ years. Gold, Silver, Goldbacks, and minted coins form the backbone of the sound money movement that spans millions of holders globally. Yet in 2026, transferring physical gold between two people still requires a bank wire, a custodian letter, or a physical shipment. There is no native digital layer.

Existing digital gold products (ETFs, allocated accounts) solve part of the problem but introduce new failure points: centralized custodians, opaque reserve reporting, restricted trading hours, and no programmability. None of them are on-chain. None of them settle in under a second. None of them can be used by a software agent or integrated into a DeFi protocol without significant engineering overhead.

No digital transfer layer	Sending physical gold requires wires, custodian letters, or physical shipment. There is no Venmo for gold.
Opaque reserve reporting	Most digital gold products publish reserve reports quarterly or annually. Verification requires trusting an auditor.
No programmability	Physical metals cannot be used in smart contracts, DeFi protocols, or agent-to-agent payments without centralized intermediaries.
Restricted access	Gold ETFs and allocated accounts have trading hours, minimum investments, and KYC barriers that exclude billions of potential holders.
Settlement latency	Cross-border precious metals settlement takes 3 to 5 business days. The metals market has not materially improved on this in decades.

2. WHY NOW

Three converging forces make 2026 the right moment for tokenized precious metals infrastructure.

Solana has reached institutional readiness

Solana now processes over 162 million transactions daily with median fees under a penny and has maintained continuous uptime since February 2024. Franklin Templeton, BlackRock, and VanEck have all tokenized flagship funds on Solana. R3, managing over \$10 billion in real-world assets, selected Solana for enterprise blockchain convergence. The institutional validation that was missing in prior cycles now exists.

The sound money movement has scale

The precious metals community represents tens of millions of holders globally. Organizations like the United Precious Metals Association, the Sound Money Trade Association, and the Goldback network have built physical distribution infrastructure over decades. This community does not lack demand for better infrastructure. It lacks a builder willing to serve it on-chain.

The agent economy needs programmable hard assets

The emergence of AI agents capable of autonomous transactions creates a new category of demand: programmable access to real assets. Coinbase revived HTTP 402 for exactly this purpose. An AI agent that can hold, transfer, and settle in physical gold via a single API call is a new primitive the market does not yet have.

3. THE SOLUTION: OBSIDIAN PROTOCOL

Obsidian Protocol is tokenized precious metals infrastructure layer built natively on Solana by BLKW3B Inc. It issues five SPL Token 2022 assets, each backed 1:1 by physical metal in insured vault custody. The protocol provides three core functions: on-chain settlement of physical metals, daily ZK-attested reserve proof, and pay-per-call API access via the x402 payment gateway.

\$510M+	60,000+	5	<400ms
Addressable TVL Day 1	Day-one user base	SPL Token 2022 assets	Settlement time

Core Protocol Functions

Token issuance	Five SPL Token 2022 assets minted on Solana, each representing a specific physical metal instrument.
Reserve attestation	Daily ZK proof published on-chain via Light Protocol confirming physical backing equals or exceeds circulating supply.
Physical redemption	Any token holder can burn tokens and receive the underlying physical metal shipped from UPMA vault custody.
Protocol fees	SPL Token 2022 transfer hooks capture a fee on every token transfer automatically at the program level.
x402 gateway	Pay-per-call API access to reserve attestation, price feeds, and mint authorization for software and AI agents.

4. WHY SOLANA

The choice of Solana is not incidental. Each core protocol function depends on capabilities that are native to Solana and either unavailable or prohibitively expensive on other chains.

SPL Token 2022 transfer hooks	Protocol fees are captured at the program level on every token transfer. No other major L1 implements this natively. On Ethereum, fee capture requires a wrapper contract with additional attack surface and gas cost.
Light Protocol ZK compression	Daily reserve attestation published on-chain costs fractions of a cent via ZK-compressed Merkle commitments. The same operation on Ethereum would cost dollars per proof.
Sub-400ms settlement	Physical metal redemption requires a settlement SLA. Solana's slot time of approximately 390ms matches the redemption SLA requirements of the AGX commercial agreement. No other L1 achieves this without an L2 trade-off.
x402 native compatibility	The HTTP 402 payment standard for AI agent pay-per-call transactions is natively compatible with Solana's transaction model. Coinbase revived this standard specifically for the Solana ecosystem.

Institutional validation

BlackRock BUIDL, Franklin Templeton FOBXX, and VanEck VBILL have all launched on Solana. R3 chose Solana for \$10B+ in RWA infrastructure. The institutional rails are already in place.

5. TOKEN ARCHITECTURE

Obsidian Protocol issues five SPL Token 2022 assets on Solana. Each token corresponds to a specific physical metal instrument held in vault custody. All five tokens were deployed on Solana devnet between April 9 and April 25, 2026.

Token	Underlying	Unit	Price Feed
xGOLD	Gold bullion	Troy oz	Spot (Pyth)
xSLVR	Silver bullion	Troy oz	Spot (Pyth)
xGLDD	Gold dollar coin	1 oz	Spot (Pyth)
xSLVD	Silver dollar coin	1 oz	Spot (Pyth)
xGLDB	Goldback note	1/1000 oz	Spot (Pyth)

SPL Token 2022 Transfer Hooks

Each Obsidian token is implemented using the SPL Token 2022 standard with transfer hook extensions enabled. On every transfer, a custom program is invoked that: (1) validates the transfer against reserve constraints, (2) captures a protocol fee in basis points, and (3) logs the transfer event for reserve accounting. This fee capture happens at the program level and cannot be bypassed by any wallet or application interacting with the token.

Token Metadata

All five tokens include on-chain metadata via the Token Metadata extension of SPL Token 2022. Metadata includes the token name, symbol, backing asset description, vault location reference, and a URI pointing to the current reserve attestation proof. This metadata is updateable only by the protocol authority key.

6. RESERVE ATTESTATION AND ZK PROOFS

The integrity of any tokenized asset protocol depends on the verifiability of its reserves. Obsidian Protocol uses a two-component attestation model: a signed daily manifest from the vault custodian, compressed into a ZK Merkle commitment via Light Protocol and published on-chain.

Step 1: Vault manifest	Daily, the vault custodian exports a signed manifest of physical reserves across all five metal categories. The manifest includes quantities, unit types, and a custodian signature. It does not reveal individual account holdings.
Step 2: Merkle commitment	The manifest is compressed into a Merkle commitment via Light Protocol's ZK compression program on Solana. The commitment is a single on-chain account that represents the entire reserve state without exposing the underlying data.
Step 3: On-chain publication	The reserve attester program publishes the Merkle root and custodian signature to the Obsidian Protocol account on Solana. The timestamp and block height are recorded. Any party can verify the commitment against the manifest hash.

Step 4: Public verification

Anyone can query the on-chain attestation account and verify that the published Merkle root matches a given manifest hash. The proof is permanent, immutable, and requires no trust in any third party.

Post-mainnet, the protocol roadmap includes upgrading to a full ZK computation proof via SP1 (Succinct Labs), which will allow the protocol to prove mathematically that total vault holdings are greater than or equal to total circulating token supply, without revealing any individual position data.

7. PHYSICAL REDEMPTION

Every Obsidian token is redeemable for its underlying physical asset. Redemption is the feature that distinguishes Obsidian Protocol from synthetic or derivative precious metals products. The physical metal is always there. The token is always redeemable.

Redemption time	3 to 5 business days from burn confirmation to physical delivery.
Redemption fee	0.25% of the redemption value, captured by the protocol.
Minimum amount	\$50 USD equivalent at current Pyth spot price.
Delivery method	Insured shipping from vault custody to holder address.
Eligible tokens	All five Obsidian tokens: xGOLD, xSLVR, xGLDD, xSLVD, xGLDB.

Redemption Flow

- Holder initiates redemption through the w3bs.fun interface or directly via the Obsidian program.
- The protocol burns the specified token amount on-chain. The burn is irreversible and publicly recorded.
- A redemption order is created in the reserve management system and forwarded to vault custody.
- The vault custodian processes the order, reduces the reserve manifest by the redeemed quantity, and ships the physical metal.
- The next daily reserve attestation reflects the updated vault balance.

8. x402 PAYMENT GATEWAY

The x402 payment gateway makes the Obsidian Protocol accessible to AI agents, software systems, and developers through a pay-per-call API model. HTTP 402 (Payment Required) was revived by Coinbase as a standard for machine-to-machine micropayments. Obsidian is the first precious metals protocol built on this standard.

Reserve attestation	/attestation	Returns the current Merkle commitment, custodian signature, and timestamp for the active reserve proof. Callable by any agent or application.
Metals price feed	/price	Returns the current Pyth Network spot price for all five metals denominated in USD. Updated at Pyth oracle frequency.
Mint authorization	/mint/authorize	Validates a mint request against current reserve capacity and returns an authorization signature for on-chain token minting.

9. THE SOUND MONEY COALITION

The Obsidian Protocol is not entering the precious metals market without existing relationships. BLKW3B has built coalition relationships with the organizations that represent the physical precious metals community. These relationships provide distribution, policy surface area, and physical infrastructure that cannot be replicated by a protocol that starts from the blockchain side alone.

AGX	The commercial custodian partner. Over 60,000 members. Physical vault custody, redemption logistics, and reserve manifest generation. The institutional backbone of the protocol.
UPMA	United Precious Metals Association. The governing body for precious metals vault standards. AGX operates within the UPMA vault network.
SMTA	Sound Money Trade Association. Policy and trade representation for the sound money industry. Active coalition relationship.
C4SM	Citizens for Sound Money. Executive Director Daniel P. Diaz. Legislative and community advocacy for sound money principles.
Goldbacks	Physical Goldback notes. The underlying asset for the xGLDB token. Goldback is the most widely distributed alternative currency backed by physical gold.

10. REVENUE MODEL

Obsidian Protocol generates revenue through three mechanisms. All three are protocol-level and do not depend on centralized intermediaries.

Transfer hook fees	A basis point fee is captured on every SPL Token 2022 transfer via the transfer hook extension. This is protocol revenue from day one of any token activity, independent of TVL growth.
Redemption fees	0.25% of every physical redemption is captured by the protocol. As circulating supply grows and redemption volume increases, this becomes a meaningful revenue stream.
x402 API fees	Pay-per-call fees on the three x402 gateway endpoints. As AI agent usage grows, this scales independently of token market cap.

11. ROADMAP

Phase	Target	Deliverables
Phase 1	Complete	Devnet deployment of all 5 SPL Token 2022 assets. Reserve attestation architecture scoped. Commercial custodian partnership initiated. Coalition relationships established.
Phase 2	May 20, 2026	Smart contract audit engagement signed. Code freeze on all 5 mints. Audit firm shortlist: OtterSec, Halborn, Zelic.
Phase 3	June 29, 2026	Mainnet deployment of all 5 tokens. Light Protocol ZK reserve attestor live and publishing daily proofs. Public storefront and reserve dashboard at w3bs.fun.

Phase 4	July 31, 2026	x402 payment gateway live across 3 endpoints. Public reserve dashboard shipping. Solana Mobile Chapter 2 integration in progress.
Phase 5	Q3-Q4 2026	SP1 ZK computation proof upgrade. Jupiter secondary market liquidity integration. Institutional onboarding. Additional metal instruments. Cross-coalition distribution expansion.

12. RISK CONSIDERATIONS

Obsidian Protocol operates at the intersection of physical asset custody, blockchain infrastructure, and financial services. The following risk categories are material to the protocol and its participants.

Custodian risk	The physical metal backing the tokens is held by a third-party vault custodian. Failure, insolvency, or misconduct by the custodian would affect the redeemability of tokens. The protocol mitigates this through daily ZK attestation and UPMA vault standards compliance.
Smart contract risk	The Obsidian Protocol programs are subject to the same risks as any on-chain code: bugs, exploits, and unexpected interactions. Mitigation: third-party audit by a recognized Solana security firm prior to mainnet.
Oracle risk	Token prices are denominated using Pyth Network price feeds. Pyth oracle manipulation or downtime could affect protocol operations. Mitigation: circuit breakers and fallback price mechanisms in the protocol design.
Regulatory risk	The regulatory treatment of tokenized precious metals varies by jurisdiction and is subject to change. BLKW3B Inc. operates as a Delaware C-Corporation and engages legal counsel on compliance matters. No representations are made regarding the regulatory status of Obsidian tokens in any jurisdiction.
Liquidity risk	Secondary market liquidity for Obsidian tokens at mainnet launch will be limited. Physical redemption provides a floor but not immediate liquidity at spot price. Jupiter integration is on the post-mainnet roadmap.

13. LEGAL DISCLAIMER

This document is provided for informational purposes only. It does not constitute an offer or solicitation to buy or sell any security, token, or financial instrument. Nothing in this document should be construed as investment advice, legal advice, or financial advice. BLKW3B Inc. makes no representations or warranties, express or implied, regarding the accuracy or completeness of the information contained herein.

Obsidian Protocol tokens are not securities, money market instruments, or investment contracts. They are digital representations of physical precious metals held in vault custody. The regulatory treatment of tokenized assets varies by jurisdiction. Prospective participants should consult qualified legal and financial advisors in their respective jurisdictions before interacting with the protocol.

This document may contain forward-looking statements regarding product development, roadmap timelines, and commercial relationships. These statements are subject to risks and uncertainties. Actual results may differ materially from those described. BLKW3B Inc. undertakes no obligation to update forward-looking statements.